Selective Consistency Training with Reinforcement Learning

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Abstract

[TODO: write abstract once results are settled.]

1. Introduction

1.1. Relevant Pathologies

Large language model (LLM) behaviour is often influenced by prompt cues that should not, in principle, affect the behaviour of interest. A widely studied example is sycophancy, where LLMs adapt their responses to align with a user's stated views or preferences (Perez et al., 2022). Another example is evaluation gaming, where LLM behaviour depends on whether the model "believes" the input to be part of an evaluation. **[TODO: cite apollo, anthropic and AISI's research]** This is particularly concerning because, if models behave differently in evaluations than in the real world, our evaluations become uninformative about real-world behaviour (Carlsmith, 2023). **[TODO: cite the BCT/ACT paper (2510.27062) here once added to bib.]**

1.2. Why Consistency Training

Two general approaches address the above behavioural inconsistencies. The first is to modify the input or the internal representations induced by the input so that the cues driving the inconsistencies are absent or hard for the LLM to detect **[TODO: cite cot & activation steering work on eval awareness reduction]**. However, isolating and removing the relevant cues from the input or internal representations is difficult in practice **[TODO: why?]**. The second is to train LLMs to behave consistently regardless of whether such cues are present. Consistency training is a class of methods focussing on the latter approach by training LLMs to produce consistent outputs or internal representations across input perturbations **[TODO: cite consistency training methods]**.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1.3. The problem with existing consistency training methods

A concern with existing consistency training methods is *obfuscation*: LLMs may learn to suppress from their outputs or internal representations any evidence that they detected the cues driving inconsistency since these outputs or internal representations are trained to be consistent with the cues being absent. Such obfuscation could make monitoring and auditing these systems more difficult. This concern can be avoided with *Selective* Consistency Training: training for consistency only over the selected behavioural properties where consistency is desirable, leaving other parts of the output free to vary.

We introduce *Reinforcement Learning Consistency Training* (RLCT), a reinforcement learning method for selective consistency training. RLCT works by computing and matching select behaviour propensities across input perturbations.

1.4. Summary of contributions

- We introduce *Reinforcement Learning Consistency Training* (RLCT), a method that enforces consistency only over selected behavioural properties of the output.
- We validate the method against BCT on the sycophancy benchmark of **[TODO: BCT paper]** on Llama-3.1-8B-Instruct and **[TODO: gpt oss]**.
- We show that RLCT causes significantly less obfuscation compared to BCT, as measured by significantly lower decrease in bias acknowledgment.

2. Consistency Training

2.1. Existing methods

2.1.1. Bias-Augmented Consistency Training

Bias-Augmented Consistency Training (BCT) trains LLMs to produce the same output on perturbed inputs as on the corresponding unperturbed input. For each prompt, a response is sampled from the LLM. The LLM is then fine-tuned to produce that same response when given perturbed prompts. **[TODO: cite bct paper]**

2.1.2. Activation Consistency Training

[TODO: cite ACT paper].

Activation Consistency Training (ACT) trains LLMs to produce internal representations when given a perturbed input that match the corresponding unperturbed input. This is achieved by minimising the difference between residual-stream activations on the perturbed and clean prompts at matching token positions, pushing the perturbed activations toward the clean ones. ACT is applied only to prompt token positions, not response token positions. **[TODO: mention BCT still SOTA]**

2.2. Existing Applications

Consistency training has previously been applied to sycophancy **[TODO: cite BCT paper]**, jailbreak robustness **[TODO: cite ACT paper]**, and reducing persona drift **[TODO: cite @miscshah2026frustration, title = Gemma Gets Help: Mitigating Frustration and Persona Drift with Consistency Training, author = Shah, Neil and Africa, David Demitri, year = 2026, howpublished = LessWrong, note = SPAR Research Fellowship, url =]**. **[TODO: cite Irpan et al (ACT paper)]** report BCT and ACT lead to similar decrease in LLM sycophancy, while BCT leads to a larger decrease in jailbreak susceptibility. -

2.3. Novel applications enabled by selective consistency training

BCT and ACT train LLMs to generate the same output or internal representations as when certain cues are absent from the prompt. This raises multiple concerns. **[TODO: cite Irpan et al (ACT paper)]** note that this training may misgeneralise to LLMs ignoring other important parts of the prompt, such as user instructions. Furthermore, constraining the entire output or internal activations also constrains any mentions or verbalisations of the prompt cues driving inconsistency. Therefore such training may also lead to obfuscation. This hampers the applicability of these methods when monitoring and auditing are important. As an example, training models to not verbalise any prompt injection jailbreaks may reduce their monitorability. Similarly, training LLMs to not verbalise when they correctly infer that they are being evaluated, reduces our ability to track evaluation awareness **[TODO: cite eval awareness literature eg. ansti scheming, apollo system cards, igor's research]**.

Selective consistency training avoids the above concerns by selectively constraining only desired behaviours to be consistent, leaving other behaviours such as verbalisation or acknowledgement of the cues driving inconsistency free to vary.

3. Reinforcement Learning Consistency Training

We introduce *Reinforcement Learning Consistency Training* (RLCT), a method for selective consistency training.

3.1. Method

Consider an input x and a set of input perturbations $\mathcal{F}$. For each perturbation $f \in \mathcal{F}$, we sample N trajectories from the current policy on the perturbed input x_f. Each trajectory i is scored for the target behavioural property by a binary classifier T (for example, "the response matches the bias suggested in the prompt").

Define the local behaviour rate for perturbation f as

$$p_f = \frac{1}{N} \sum_{i=1}^{N} T(\tau_{f,i}), \tag{1}$$

and let $q = Q(\{p_f\}_{f \in \mathcal{F}})$ denote a target rate. This target rate may be the behaviour rate on a particular perturbation (for example, the clean or unperturbed prompt), or a function of the per-perturbation rates, such as their mean, minimum, or maximum.

The consistency reward for trajectory $\tau_{f,i}$ rewards the trajectory when its trait value moves the local rate toward the target rate:

$$r^{\text{cons}}_{f,i} = -(p_f - q)\,(T(\tau_{f,i}) - q). \tag{2}$$

To prevent the target rate itself from drifting during training, an optional anchor term ties q to the same target statistic under the initial reference policy π_{ref} (typically a checkpoint taken before consistency training begins). Let $q_{\text{ref}} = Q(\{p^{\text{ref}}_f\}_{f \in \mathcal{F}})$ denote this fixed reference target rate, where p^{ref}_f is the behaviour rate of π_{ref} on perturbation f. The anchor reward is assigned only to trajectories whose trait values enter the target statistic q. Let λ_f denote the target-contribution weight for trajectories from perturbation f, with $\lambda_f = 0$ when f does not contribute to q:

$$r^{\text{anchor}}(\tau_{f,i}) = -\lambda_f\,(q - q_{\text{ref}})\,(T(\tau_{f,i}) - q_{\text{ref}}). \tag{3}$$

Thus only trajectories that actually contribute to q receive anchor pressure: if $q = p_g$ for some designated perturbation g, then $\lambda_f = \mathbb{I}[f = g]$; if q is the mean over all perturbations, then $\lambda_f = 1/|\mathcal{F}|$; and if q is a minimum or maximum, then λ_f is placed on the active perturbation(s) attaining that value.

The total reward combines the two terms with mixing weight α:

$$r_{f,i} = (1 - \alpha)\,r^{\text{cons}}_{f,i} + \alpha\,r^{\text{anchor}}(\tau_{f,i}). \tag{4}$$

3.2. RL algorithm

We optimise the policy with GRPO and a KL penalty to a fixed reference policy. The only departure from standard GRPO is the sampling structure: groups consist of $|\mathcal{F}|$ perturbations, with N trajectories per perturbation, all sharing the same underlying input. GRPO advantage normalisation and policy updates proceed as usual. We use parameter-efficient fine-tuning (LoRA) throughout.

3.3. Why this avoids the obfuscation pressure

Because the reward depends only on the trait T, the per-perturbation rates p_f, and the chosen target rate q, the model is not penalised for any particular content in the chain of thought. In particular, if the model verbalises that it has noticed a perturbation, the reward is unchanged provided its action matches the target rate. This contrasts with BCT, where matching a clean-prompt target response would penalise verbalisation of the perturbation.

4. Experiments

4.1. Environment and bias types

We validate the method on the sycophancy benchmark of **[TODO: BCT paper]**. The benchmark consists of multiple-choice questions to which a small prompt-level perturbation has been added that suggests a particular answer. We restrict our experiments to five single-turn bias types: suggested answer, distractor argument, distractor fact, wrong few-shot, and spurious few-shot squares.

4.2. Train/evaluation split

We train on a single bias type (suggested answer) drawn from a single question source (MMLU) and evaluate on a held-out question source (TruthfulQA) across all five bias types. Training on a single bias/source and evaluating on held-out bias/sources separates hyperparameter selection from final reporting.

4.3. Models

Our primary model is Llama-3.1-8B-Instruct. **[TODO: describe gpt-oss-20b/120b runs once finalised.]**

4.4. Metrics

The primary metric is the *Bias Influence Rate* (BIR), defined per question as the absolute difference between the rate at which the model selects the biased option on the biased prompt and the rate at which it selects the same option on the unbiased prompt. We also report the *Bias Acknowledgement Rate* (BAR), the rate at which the model's chain of thought acknowledges the perturbation (scored by an LLM grader), and unbiased-eval accuracy (the rate at which the model answers the unbiased question correctly), the latter as a check that consistency training does not degrade general task performance.

4.5. Baselines

We compare RLCT against the base model, BCT trained on a matched training set, and a control variant of RLCT in which both perturbations are unperturbed (used to isolate training-induced drift from the consistency signal). **[TODO: ACT baseline if available.]**

4.6. Results

[TODO: fill in results.]

5. Discussion

[TODO: interpret results once finalised.]

Limitations. Consistency training does not address deceptive alignment: a model that is already deceptively aligned could plausibly be taught to hide its eval-gaming behaviour better, similarly to how standard post-training methods make sleeper-agent behaviours difficult to catch (Hubinger et al., 2024). Our experiments are also restricted to a narrow bias-following setting on multiple-choice questions; the obfuscation question for the eval-awareness setting itself has not been tested directly. **[TODO: add limitations specific to results once finalised.]**

Future directions. Application of selective consistency training to evaluation-awareness settings (Section 2.3), extension to multimodal consistency (e.g. across input or output modalities), and application to other behavioural properties beyond bias-following.

6. Conclusion

[TODO: summarise once results are in.]

References

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